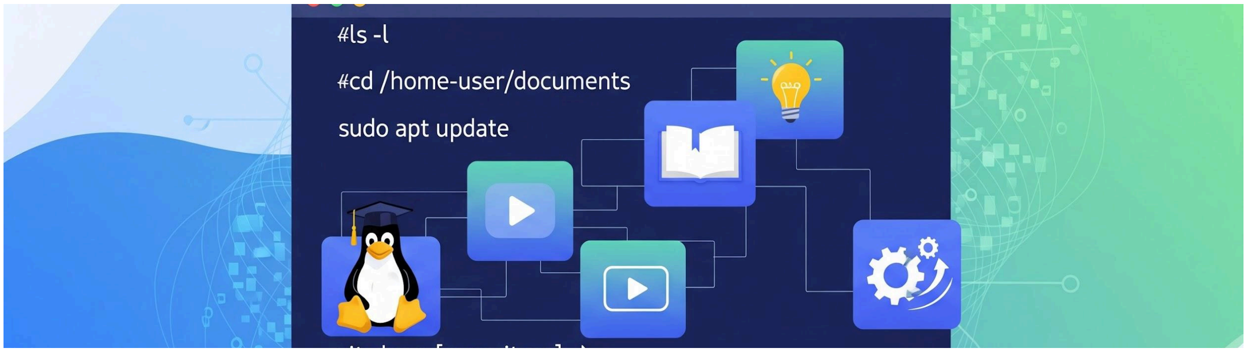


LINUX OS

Introduction:

- Linux is a free and open-source operating system with high security.
- Linux is a multi-user operating system.

OPERATING SYSTEM

- An Operating System (OS) is software that acts as an interface between computer hardware components and the user.
- Every computer system must have at least one operating system to run other programs.
- Applications like browsers, MS Office, Notepad, games, etc., need an environment to run and perform tasks.
- The OS helps users communicate with the computer without knowing the computer's language.

KERNEL

- The kernel manages the hardware components such as CPU, memory, and peripheral devices.
- The kernel is the lowest level of the operating system.

DAEMONS

- Daemons manage background services such as printing, sound, scheduling, etc.
- These services start during system boot or after user login.

SHELL

- The shell is an environment where users can run commands, programs, and shell scripts.
- It takes input from the user, executes it, and displays the output.
- A shell acts as an interface between the user and the operating system.

Shells can be:

Ø Graphical (GUI-based)

Ø Command-line based (CLI)

COMMAND

· A command is an instruction or request given to the operating system by a user.

· It tells the computer to perform a specific task.

TERMINAL

· A terminal is a text-based interface that allows users to interact with the operating system by typing commands and receiving output.

· Terminals are commonly used in Linux, Unix, macOS, and Windows systems.

LINUX OS DISTRIBUTIONS

· Linux distributions are modified versions of Linux released with different names based on user requirements.

Examples:

i. RedHat

ii. Ubuntu

iii. Debian

iv. CentOS

v. Fedora

vi. OpenSUSE

vii. Kali Linux

viii. Amazon Linux

ix. Rocky Linux

LINUX HISTORY

- In 1991, Linus Torvalds, a student at the University of Helsinki, Finland, started developing Linux.
- The Linux kernel is written in C language.
- Initially, the name was planned as Freax, but later it became Linux.
- In 1992, Linux was released under the GNU General Public License.

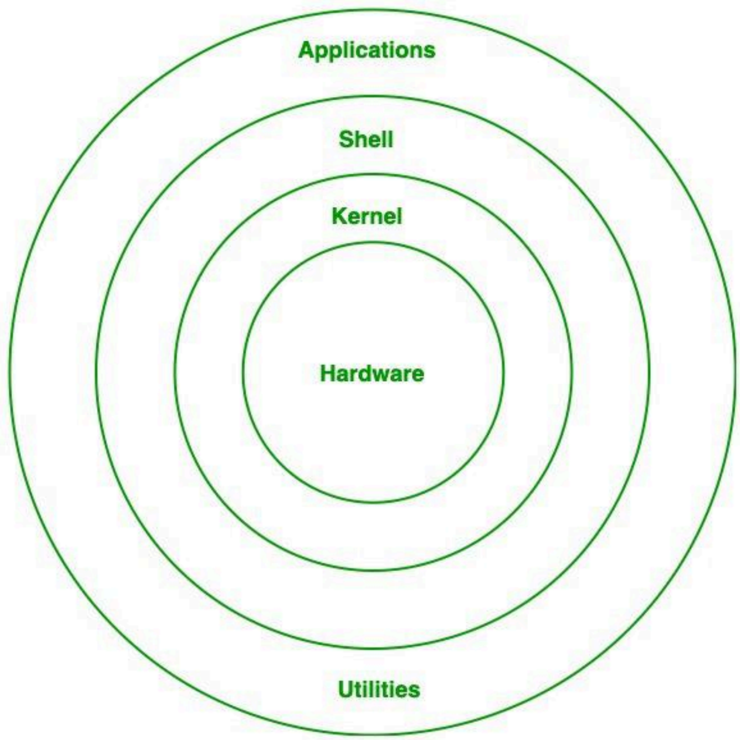
OPEN SOURCE

- Linux is distributed under an open-source license.
- Open source provides:
- Freedom to use the software
- Freedom to study and modify the code
- Freedom to redistribute copies
- Freedom to distribute modified versions

ARCHITECTURE

The Linux architecture consists of:

- Kernel
- Shell
- Commands
- Terminal



Introduction to Essential Linux Commands

Linux is a powerful and popular operating system, especially for servers, developers, and advanced users. To effectively use Linux, you interact with it through the **Command Line Interface (CLI)**, also known as the terminal or shell.

This document provides a detailed, yet simple, guide to essential Linux commands.

Basic Navigation and File Management

These commands help you move around the filesystem and handle files and directories (folders).

Command	Simple Language Explanation	Example
`pwd`	**P**rint **W**orking **D**irectory. Shows you the full path of the directory you are currently in.	`pwd`
`ls`	**Lis**t. Shows you the files and folders inside your current directory.	`ls`
`ls -l`	A "long list" that shows detailed information (permissions, size, owner, date).	`ls -l`
`cd`	**C**hange **D**irectory. Used to move from one directory to another.	`cd Documents` (moves into the Documents folder)
`cd ..`	Moves you up one level (to the parent directory).	`cd ..`
`cd ~`	Moves you directly to your user's Home directory (the starting point).	`cd ~`
`mkdir`	**M**ake **Dir**ectory. Creates a new folder.	`mkdir new_project`
`touch`	Creates a brand-new, empty file.	`touch notes.txt`

```
ubuntu:~$ cd app
bash: cd: app: Not a directory
ubuntu:~$ cd data
bash: cd: data: Not a directory
ubuntu:~$ ls
a.txt  app  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ cd a
a.txt  app
ubuntu:~$ cd a
a.txt  app
ubuntu:~$ cd app
bash: cd: app: Not a directory
ubuntu:~$ touch app
ubuntu:~$ ls
a.txt  app  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ mkdir app
mkdir: cannot create directory 'app': File exists
ubuntu:~$ mkdir app2
ubuntu:~$ cd app2
ubuntu:~/app2$ cd ..
ubuntu:~$ rmdir app
rmdir: failed to remove 'app': Not a directory
ubuntu:~$ ls
a.txt  app  app2  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ nono app
Command 'nono' not found, did you mean:
  command 'mono' from deb mono-runtime (6.8.0.105+dfsg-3.5ubuntu1)
  command 'nano' from deb nano (7.2-2ubuntu0.1)
  command 'nona' from deb hugin-tools (2023.0.0+dfsg-1)
Try: apt install <deb name>
ubuntu:~$ cd app2
ubuntu:~/app2$ nano file
ubuntu:~/app2$ ls
file
ubuntu:~/app2$ cat file
Hello Linux
This is my first file
ubuntu:~/app2$ vi file
ubuntu:~/app2$ cat file
Hello Linux
This is my first file
ok new line added here ok
```

File Operations

Once you have files, you need commands to view, copy, move, and delete them.

Command	Simple Language Explanation	Example
cat	**Cat**enate. Shows the entire content of a file directly in the terminal.	cat notes.txt
less	Views file content page-by-page, good for large files. Press q to quit.	less very_large_log.txt
cp	**Cop**y. Copies a file from one location to another.	cp file1.txt /home//backup/
mv	**M**o**v**e. Moves a file (like cutting and pasting). Also used to rename files.	mv oldname.txt newname.txt
rm	**Rem**ove. Permanently deletes a file (be careful, there is no Recycle Bin).	rm temp.log
rm -r	Removes an empty or non-empty directory. The -r stands for recursive.	rm -r old_folder

Getting Help

The Linux command line has built-in help features for almost every command.

Command	Simple Language Explanation	Example
man	**Man**ual. Shows the comprehensive reference manual page for a command.	man ls

Command	Simple Language Explanation	Example
`--help`	A simple option available on most commands that provides a quick list of options.	`ls --help`

Viewing System Information

These commands provide insights into how your system is running.

Command	Simple Language Explanation	Example
`date`	Displays the current system date and time.	`date`
`whoami`	Tells you the name of the user you are currently logged in as.	`whoami`
`uname -a`	Displays detailed information about your operating system kernel.	`uname -a`
`df -h`	**D**isk **F**ree. Shows the amount of available disk space, in "human-readable" format.	`df -h`
`top`	Shows a real-time list of running processes, like the Windows Task Manager. Press `q` to quit.	`top`

Search and Location

Commands to find files or text within files.

Command	Simple Language Explanation	Example
`grep`	**G**lobal **R**egular **E**xpression **P**rint. Searches for specific text inside files.	`grep "error" server.log` (finds the word "error" in the log file)
`find`	Searches for files and directories based on various criteria (name, size, date).	`find . -name "*.txt"` (finds all files ending in `.txt` in the current directory)

Permissions

Linux uses permissions to control who can read, write, or execute a file.

Command	Simple Language Explanation	Example
`chmod`	**Ch**ange **Mod**e. Changes the permissions of a file or folder.	`chmod +x script.sh` (makes a file executable)
`chown`	**Ch**ange **Own**er. Changes the owner of a file or folder.	chown :users important_file.conf

---

LINUX COMMANDS

1.ls

What: ls is a command used to display the list of files and folders in a directory.
Why: It helps you quickly see what contents are available in the current location.

```
controlplane:~$ ls
file1  file2  file3  filesystem  folder1  folder2
```

2.touch

What: touch is a command used to create a new empty file or update the timestamp of an existing file.
Why: It helps you quickly make files or refresh file modification time.

```
controlplane:~$ touch file1 file2 file3
```

3.mkdir

What: mkdir is a command used to create a new directory (folder).
Why: It helps you organize files by making separate folders for storage.

```
controlplane:~$ mkdir folder1 folder2
```

4.pwd

What: pwd is a command used to show the full path of the current working directory.
Why: It helps you know exactly where you are in the file system.

```
controlplane:~/folder1$ pwd
/root/folder1
```

5.cd

What: cd is a command used to change the current directory.
Why: It helps you move between folders to access different files and locations.

```
controlplane:~$ cd folder1
controlplane:~/folder1$ cd folder2/folder3
controlplane:~/folder1/folder2/folder3$ cd ..
controlplane:~/folder1/folder2$ cd folder3
controlplane:~/folder1/folder2/folder3$ cd ../..
controlplane:~/folder1$
```

6.vi

What: vi is a command-line text editor used to create and edit files.
Why: It helps you write or modify file content directly from the terminal.

```
controlplane:~$ vi file1
```

- **i ->What:** i is a key in **vi editor** used to enter *Insert mode* for typing text.
  Why: It allows you to start editing and adding content inside the file.
- **Esc->What:** Esc is a key in vi used to exit Insert mode and return to Command mode.
  Why: It helps you stop typing and run vi commands like save or quit.
- **:wq->What:** :wq is a vi command used to save the file and exit the editor.
  Why: It helps you store changes and close vi safely.

7.cat

What: cat is a command used to display the content of a file on the terminal.
Why: It helps you quickly view or check what is inside a file.

```
controlplane:~$ cat file1
hiii
```

8.nano

What: nano is a simple command-line text editor used to create and edit files.
Why: It allows easy editing of files directly in the terminal with on-screen instructions.

```
controlplane:~$ nano file1
controlplane:~$ cat f1
oohiii
```

changed file content and name to f1

9.rm

What: rm is a command used to delete files or directories.
Why: It helps you remove unwanted files or clean up space.

```
controlplane:~$ ls
f1  file2  file3  filesystem  folder1  folder2
controlplane:~$ rm file2 file3 folder1
rm: cannot remove 'folder1': Is a directory
controlplane:~$ ls
f1  filesystem  folder1  folder2
```

- **Rm -rf <filename>** -> this will delete folder or file even if folder is not empty

```
controlplane:~$ ls
f1  filesystem  folder1  folder2
controlplane:~$ rm -rf folder1
controlplane:~$ ls
f1  filesystem  folder2
```

10.rmdir

What: rmdir is a command used to delete an empty directory.
Why: It helps you remove unnecessary folders from the system.

```
controlplane:~$ ls
f1  filesystem  folder2
controlplane:~$ rmdir folder2
controlplane:~$ ls
f1  filesystem
```

11.history

What: history is a command used to show a list of previously executed commands.
Why: It helps you review or reuse past commands easily.

```
controlplane:~$ history
    1  exit
    2  halt
    3  FILE=/ks/wait-background.sh; while ! test -f ${FILE}; do clear; sleep 0.01; done; bash ${FILE}
    4  ls
    5  mkdir folder1
    6  touch file1
    7  history
```

12.cal

What: cal is a command used to display a calendar in the terminal.
Why: It helps you quickly check dates without leaving the command line.

```
controlplane:~$ cal
    January 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

13.df

What: df is a command used to display the available and used disk space on file systems.
Why: It helps you check storage usage and manage disk space.

```
controlplane:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
tmpfs             230020     2568    227452   2% /run
/dev/vda1       19221248 10443404   8761460  55% /
tmpfs            1150084       84   1150000   1% /dev/shm
tmpfs               5120        0      5120   0% /run/lock
/dev/vda16        901520   119048    719344  15% /boot
/dev/vda15        106832     6250    100582   6% /boot/efi
tmpfs             230016       96    229920   1% /run/user/112
shm                65536        0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/77d577263693bd800
fde271156bf162ec69379f07e0d951d0c0f705496d46441/shm
shm                65536        0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/c8589c56b457447fc
24285c71e7bea78e560dd3e76b2958b470ef04a31e34a72/shm
shm                65536        0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/f3b4bf9e9eee7fe62
17a0f23e19d6f3431f891b0b3fb1a1be843db768b72a6d8/shm
shm                65536        0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/5710489d7e89dc4a9
7ced2b10d6e8b2250830fc013d7db41a96d09f2fc5ced56/shm
overlay         19221248 10443404   8761460  55% /run/containerd/io.containerd.runtime.v2.task/k8s.io/77d577263693bd80
0fde271156bf162ec69379f07e0d951d0c0f705496d46441/rootfs
```

14.whoami

What: whoami is a command used to display the current logged-in user.
Why: It helps you confirm which user account is active in the session.

```
controlplane:~$ whoami
root
```

15.ping

What: ping is a command used to check the network connectivity to a server or IP address.
Why: It helps you verify if a device or website is reachable and measure response time.

```
controlplane:~$ ping www.google.com
PING www.google.com (172.217.20.36) 56(84) bytes of data.
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=1 ttl=115 time=12.9 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=2 ttl=115 time=13.2 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=3 ttl=115 time=13.1 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=4 ttl=115 time=12.7 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=5 ttl=115 time=12.5 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=6 ttl=115 time=12.7 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=7 ttl=115 time=12.8 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=8 ttl=115 time=12.7 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=9 ttl=115 time=12.9 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=10 ttl=115 time=13.1 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=11 ttl=115 time=12.8 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=12 ttl=115 time=13.0 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=13 ttl=115 time=12.9 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=14 ttl=115 time=12.8 ms
```

16.mv

What: mv is a command used to move or rename files and directories.
Why: It helps you organize files by relocating them or changing their names.

```
controlplane:~$ mv file1 omkar1
controlplane:~$ ls
filesystem  folder1  omkar1
```

```
controlplane:~$ ls
filesystem  folder1  omkar1
controlplane:~$ mv omkar1 folder1
controlplane:~$ ls
filesystem  folder1
controlplane:~$ cd folder1
controlplane:~/folder1$ ls
omkar1
```

17.cp

What: cp is a command used to copy files or directories.
Why: It helps you create duplicates for backup or use in different locations.

```
controlplane:~/folder1$ cp omkar1 omkar2
controlplane:~/folder1$ ls
omkar1  omkar2
```

18.wget

What: wget is a command used to download files from the internet using a URL.
Why: It helps you quickly fetch web content or software directly from the terminal.

```
controlplane:~/folder1$ wget https://downloads.apache.org/tomcat/tomcat-11/v11.0.18/bin/apache-tomcat-11.0.18.exe
--2026-01-30 06:33:58--  https://downloads.apache.org/tomcat/tomcat-11/v11.0.18/bin/apache-tomcat-11.0.18.exe
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.214.104, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14817720 (14M) [application/x-msdos-program]
Saving to: 'apache-tomcat-11.0.18.exe'

apache-tomcat-11.0.18.exe     100%[=============================================>]  14.13M  6.36MB/s    in 2.2s

2026-01-30 06:34:01 (6.36 MB/s) - 'apache-tomcat-11.0.18.exe' saved [14817720/14817720]
```